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MILITARY SPECIFICATION

TEST SET, ELECTRICAL, TORPEDO CONTROL CABLE, MARK 519 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers requirements for the procurement of Torpedo Control Cable Electrical Test Set Mk 519 Mod 0, DL 2837964. The test set is used to inspect and test torpedo control cables in heated work areas.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-16400	Electronic Equipment, Naval Ship and Shore: General Specification
MIL-T-18404	Torpedoes, Environmental Requirements, General Specification for
MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for

FSC 6625

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-type base
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

PUBLICATIONS

Naval Ordnance Systems Command

OP 2639	Torpedo Control Cables and Test Equipment

DRAWINGS

Naval Ordnance Systems Command

LD 161783	Torpedo Control Cable Mk 5 Mod 1 Test Control Unit
LD 162042	Torpedo Control Cable Mk 3 Mod 13 Submarine Unit Assembly
LD 162107	Torpedo Control Cable Mk 3 Mod 14
LD 162420	Torpedo Control Cable Mk 6 Mod 0
LD 162531	Torpedo Control Cable Assembly Mk 7 Mod 0
LD 162533	Torpedo Control Cable Assembly Mk 7 Mod 1
LD 162596	Torpedo Connector Receptacle Assembly
LD 162737	Simulator Cable Mk 19 Mod 0

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LD 162838	Connector, Torpedo Tube Cable Mk 42 Mod 1
LD 162872	Cable Assembly, Missile Control Mk 8 Mod 0
LD 162873	Cable Assembly, Missile Control Mk 9 Mod 0
LD 275029	Torpedo Control Cable Mk 2 Mod 1 A. W. Tube Unit
LD 287295	Torpedo Control Cable Mk 3 Mod 7 SUB Tube Unit
LD 411098	Torpedo Control Cable Mk 1 Mod 12 Torpedo Unit
LD 411099	Torpedo Control Cable Mk 1 Mod 13 Torpedo Unit
LD 414200	Torpedo Control Cable Mk 3 Mod 10 Submarine Unit Assembly
LD 414201	Torpedo Control Cable Mk 3 Mod 9 Submarine Unit
LD 414218	Torpedo Control Cable Mk 3 Mod 12 Submarine Unit Assembly
DL 2837964	Test Set Mk 519 Mod 0 Assembly and Accessory Kit
Dwg 2837970	Shorting Adapter Assembly
Dwg 2837984	Calibration Test Adapter Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Torpedo Control Cable Electrical Test Set Mk 519 Mod 0 is a portable self-contained unit for use in heated work areas to test a variety of multiconductor cable assemblies for continuity and insulation resistance.

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3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the cable test set shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing test sets that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the test set shall conform strictly to the detailed requirements indicated on the drawings and specifications in DL 2837964.

3.2.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the test set shall be strictly in accordance with the applicable drawings, specifications, and publications listed in DL 2837964.

3.2.4 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in DL 2837964.

3.2.5 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Performance requirements and product characteristics. The test set shall meet the following performance requirements and product characteristics:

3.3.1 Product characteristics

3.3.1.1 Mating. The test set shall be capable of mating with the following cables and connectors:

Cable	Mod	LD
Mk 1	12	411098
Mk 1	13	411099
Mk 2	1	275029
Mk 3	7	287295
Mk 3	9	414201
Mk 3	10	414200
Mk 3	12	414218
Mk 3	13	162042
Mk 3	14	162107

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Cable	Mod	LD
Mk 6	0	162420
Mk 7	0	162531
Mk 7	1	162533
Mk 19	0	162737
Mk 5	1	161783
Mk 8	0	162872
Mk 9	0	162873

Connector	Mod	LD
Mk 42	1	162838
Torpedo Receptacle	-	162596

3.3.1.2 Power. The test set shall operate when energized by 115 volts ±10 percent 60 Hertz (Hz) single phase source. Connection to the source shall be through a standard 3-wire grounded plug.

3.3.1.3 Fault indication. Faults shall be indicated by a system of lights related to a chart for conductor identification.

3.3.2 Performance requirements

3.3.2.1 Operation. The test set shall be capable of both an automatic and a manual mode of operation, with manual selection of the mode via a front panel switch.

3.3.2.1.1 Automatic mode. In the automatic mode, the test set shall sequentially test each conductor for the selected attributes (continuity or insulation). Each conductor shall be tested for a minimum of 5 seconds, and the complete sequence of 65 conductors shall be completed within a maximum time of 20 minutes. Sequencing shall stop in the event of a conductor failure. The test set shall be capable of clearing faults by manually by-passing the faulty conductor before returning to the automatic mode of operation.

3.3.2.1.2 Manual mode. In the manual mode, the test set shall be sequenced through the 65 conductors by means of a manually operated control on the front panel. The tests performed shall be the same as in the automatic mode.

3.3.2.2 Continuity test voltage. The test set shall be capable of detecting faults 10 percent or more above the nominal value selected on the panel when tested at a potential of 4-6 vdc applied between the conductor under test and all others tied together. The required shorting adapter, as supplied with the test set, shall be installed on the free end of the unit under test.

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3.3.2.3 Insulation resistance test voltage. The test set shall be capable of detecting faults 10 percent or more below the nominal value selected on the panel when tested at a potential of 725 ±25 vdc applied between the conductor under test and all others tied together.

3.3.2.4 Insulation resistance. The insulation resistance of the test set circuitry shall be such that a current flow of less than 5 microamperes shall occur when the circuitry is energized at 1000 volts dc in accordance with the tests of 4.6.5.

3.4 Environmental requirements. The test set shall be subjected to the following environmental conditions and during or thereafter, as applicable, meet all requirements specified herein:

3.4.1 Temperature, operating. The test set shall be capable of operating throughout the temperature range specified in 4.6.6 and thereafter shall meet all performance requirements and product characteristics of this specification.

3.4.2 Temperature, nonoperating. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the nonoperating temperature tests of 4.6.7.

3.4.3 Shock. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the shock test of 4.6.8.

3.4.4 Vibration, operating. The test set shall be subjected to the operational vibration test of 4.6.9 and during and thereafter shall meet all performance requirements and product characteristics of this specification.

3.4.5 Vibration, transportation. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the vibration, transportation test of 4.6.10.

3.4.6 Marine environment. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the marine environment test of 4.6.11.

3.4.7 Temperature and humidity cycling. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the temperature and humidity cycling tests of 4.6.12.

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3.4.8 Tropical environment. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the tropical environment test of 4.6.13.

3.4.9 Accelerated life. The test set shall be capable of meeting all performance requirements and product characteristics of this specification after being subjected to the accelerated life test of 4.6.14.

3.5 Workmanship. The test sets furnished to this specification shall be constructed and finished in a manner that will ensure full compliance with all requirements of this specification and applicable NAVORD drawings. Particular attention shall be paid to quality of soldering and wiring and to freedom from defects and burrs or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the test set shall be classified as follows:

(a) Preproduction inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

4.3 Preproduction

4.3.1 Preproduction sample. A preproduction sample of two test sets, manufactured in accordance with 3.2.1, shall be subjected to the preproduction tests specified in Table I at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test set by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

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TABLE I. PREPRODUCTION TEST SEQUENCE

Examination or Test	Requirement Paragraph	Method Paragraph	Sample Numbers
Examination of Product	3.2, 3.5	4.6.1	1, 2
Operation	3.3.2.1	4.6.2	1, 2
Continuity Test Voltage	3.3.2.2	4.6.3	1, 2
Insulation Resistance Test Voltage	3.3.2.3	4.6.4	1, 2
Insulation Resistance	3.3.2.4	4.6.5	1, 2
Temperature, Operating	3.4.1	4.6.6	1
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Temperature, Nonoperating	3.4.2	4.6.7	2
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Shock	3.4.3	4.6.8	1, 2
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Vibration, Operating	3.4.4	4.6.9	1, 2
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Vibration, Transportation	3.4.5	4.6.10	2
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Marine Environment	3.4.6	4.6.11	1
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Temperature and Humidity Cycling	3.4.7	4.6.12	2
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Tropical Environment	3.4.8	4.6.13	1
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	
Accelerated Life	3.4.9	4.6.14	1
Operation	3.3.2.1	4.6.2	
Insulation Resistance	3.3.2.4	4.6.5	

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4.3.2 Preproduction test sequence. The tests defined in 4.6 shall be performed in the sequence shown in Table I. No deviation from the test sequence shall be permitted without prior approval of the procuring activity. When specified in the contract or purchase order, certified inspection data covering the results of preproduction inspections shall be forwarded to the procuring activity.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

(a) Individual inspection

(b) Sampling plans and tests

4.4.1 Individual inspection. The manufacturer shall maintain a 100 percent inspection procedure to ensure that each test set meets the requirements of Table II.

TABLE II. INDIVIDUAL INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.2, 3.5	4.6.1
Operation	3.3.2.1	4.6.2
Insulation Resistance	3.3.2.4	4.6.5

4.4.2 Sampling plan and tests. Test sets shall be selected in accordance with MIL-STD-105, Level II at an AQL of 0.95 percent. The AQL applies to the characteristics verified as a group.

4.4.2.1 Lot. A lot shall consist of a continuous run of test sets produced by the same process with the same equipment in a quantity not greater than 50 units.

4.4.2.2 Lot acceptance tests. Lot acceptance tests shall be as shown in Table III.

4.5 Test conditions. Unless otherwise specified, the tests and examinations shall be conducted under any combination of conditions within the ranges below. Any specified condition shall not affect the other two ambient ranges.

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(a) Temperature: 20 degrees to 30 degrees centigrade (C)

(b) Relative humidity: Room ambient to 95 percent

(c) Barometric pressures: 24 to 31 inches of mercury

TABLE III. LOT ACCEPTANCE TESTS

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.2, 3.5	4.6.1
Performance Requirements	3.3	4.6.2, 4.6.3, 4.6.4, 4.6.5

4.6 Inspection procedures

4.6.1 Examination of product. The test sets shall be examined to ensure conformance to the applicable NAVORD drawings and to the requirements in (a) and (b) below. Assurance shall be made that no changes have been made in the test sets other than those granted for preproduction approval.

(a) Materials, design, and construction (3.2)

(b) Workmanship (3.5)

4.6.2 Operation. Each test set shall operate satisfactorily when operated in accordance with NAVORD OP 2639 and tested as below:

4.6.2.1 Automatic mode. The test set shall step through all 65 conductors to the home position within 20 minutes when operating with Shorting Adapter Assembly "A" (Dwg 2837970) installed and while in the 0.5 ohm continuity test configuration. There shall be a minimum of 5 seconds between successive steps.

4.6.2.2 Manual mode. With Calibration Test Adapter Assembly (Dwg 2837984) installed and the test set configured as required under the test select column of Table IV, the test set shall be manually stepped through each conductor and the required indication as listed in Table IV shall be obtained. There shall be a minimum of 5 seconds between successive steps, and the fault indication system shall operate where appropriate.

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TABLE IV

CALIBRATION TEST DATA

Step	P1 Pin	Test Select		Required Indication	
		Function	Test	Pass/Fault	Wire
1	A	Continuity	1.0 Ohm	Pass	00
2	B	Continuity	1.0 Ohm	Fault	01
3	C	Continuity	1.0 Ohm	Pass	02
4	D	Continuity	1.0 Ohm	Fault	03
5	E	Continuity	1.0 Ohm	Pass	04
6	F	Continuity	1.0 Ohm	Fault	05
7	G	Continuity	1.0 Ohm	Pass	06
8	H	Continuity	1.0 Ohm	Pass	07
9	I	Continuity	1.0 Ohm	Pass	08
10	J	Continuity	1.0 Ohm	Fault	09
11	K	Continuity	1.0 Ohm	Pass	10
12	L	Continuity	1.0 Ohm	Pass	11
13	M	Continuity	1.0 Ohm	Pass	12
14	N	Continuity	1.0 Ohm	Fault	13
15	O	Continuity	1.0 Ohm	Fault	14
16	P	Continuity	0.5 Ohm	Pass	15
17	Q	Continuity	0.5 Ohm	Fault	16
18	R	Continuity	0.5 Ohm	Pass	17
19	S	Continuity	0.5 Ohm	Fault	18
20	T	Continuity	0.5 Ohm	Pass	19
21	U	Continuity	0.5 Ohm	Fault	20
22	V	Continuity	0.5 Ohm	Pass	21
23	W	Continuity	0.5 Ohm	Pass	22
24	X	Continuity	0.5 Ohm	Pass	23
25	Y	Continuity	0.5 Ohm	Pass	24
26	Z	Continuity	0.5 Ohm	Pass	25
27	&	Continuity	0.5 Ohm	Pass	26
28	a	Continuity	0.5 Ohm	Fault	27
29	b	Continuity	0.5 Ohm	Fault	28
30	c	Continuity	0.5 Ohm	Fault	29
31	d	Continuity	0.5 Ohm	Fault	30
32	e	Continuity	0.5 Ohm	Fault	31
33	f	Continuity	0.5 Ohm	Fault	32
34	g	Leakage	1.0 Meg	Pass	33
35	h	Leakage	1.0 Meg	Fault	34
36	i	Leakage	1.0 Meg	Pass	35
37	j	Leakage	1.0 Meg	Fault	36
38	k	Leakage	1.0 Meg	Pass	37
39	l	Leakage	1.0 Meg	Pass	38

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TABLE IV

CALIBRATION TEST DATA (Cont'd)

Step	P1 Pin	Test Select		Required Indication	
		Function	Test	Pass/Fault	Wire
40	m	Leakage	1.0 Meg	Pass	39
41	n	Leakage	1.0 Meg	Fault	40
42	o	Leakage	1.0 Meg	Fault	41
43	p	Leakage	1.0 Meg	Pass	42
44	q	Leakage	1.0 Meg	Pass	43
45	r	Leakage	1.0 Meg	Pass	44
46	s	Leakage	1.0 Meg	Pass	45
47	t	Leakage	70 Meg	Pass	46
48	u	Leakage	70 Meg	Fault	47
49	v	Leakage	70 Meg	Pass	48
50	w	Leakage	70 Meg	Fault	49
51	x	Leakage	70 Meg	Pass	50
52	y	Leakage	70 Meg	Pass	51
53	z	Leakage	70 Meg	Pass	52
54	AA	Leakage	70 Meg	Pass	53
55	AB	Leakage	70 Meg	Pass	54
56	AC	Leakage	70 Meg	Pass	55
57	AD	Leakage	70 Meg	Pass	56
58	AE	Leakage	70 Meg	Pass	57
59	AF	Leakage	70 Meg	Pass	58
60	AG	Leakage	70 Meg	Fault	59
61	AH	Leakage	70 Meg	Fault	60

4.6.3 Continuity test voltage. The test set shall be configured for the 1.0 ohm continuity check and manually operated through the 65 conductors. A voltage reading shall be taken at each step between the conductor under test and any other conductor. The readings shall fall within the range of 4-6 vdc.

4.6.4 Insulation resistance test voltage. The test set shall be configured for the 1.0 megohm insulation resistance check and manually operated through the 65 conductors. A voltage reading shall be taken at each step between the conductor under test and any other conductor. The readings shall fall within the range 725 ± 25 vdc.

4.6.5 Insulation resistance. A 1000 ± 25 volt dc potential shall be applied between two points, one point being each circuit in turn, and the other point all other circuits and the carrying case tied together to

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ground. The potential shall be applied for 1 minute minimum (to a maximum of 2 minutes) to attain a stable reading. The insulation resistance shall exceed 200 megohms.

4.6.6 Temperature, operating. The test set shall be capable of normal operation (without alignment or adjustment, other than of accessible controls employed for operation of the equipment) throughout the following temperature cycle. At the conclusion of this test the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

(a) Hold temperature at 0 degrees ± 2 degrees C for at least 24 hours.

(b) Increase temperature in steps of 10 degrees C each, at 30 minutes per step, until +50 degrees ± 2 degrees C is reached, and hold at that temperature for at least 4 hours.

(c) Reduce room temperature in steps of 10 degrees C, at 30 minutes per step, until +25 degrees C ± 2 degrees C is reached, and hold at that temperature for at least 4 hours.

4.6.7 Temperature, nonoperating. The test set shall be capable of normal operation immediately after passing through the following temperature cycle in the stowed condition. At the conclusion of this test the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

(a) With the equipment conditioned at room temperature of 25 degrees ± 2 degrees C, reduce the temperature to -62 degrees C and hold to within +0 degrees -5 degrees C of that temperature for at least 3 days.

(b) Increase the temperature to +75 degrees C and maintain that temperature within ± 3 degrees C for at least 4 hours.

(c) Reduce temperature to +25 degrees ± 2 degrees C and maintain at that temperature for at least 4 hours.

The recommended rate of temperature change shall be in approximately 10 degrees C steps, at 30 minutes per step, so that thermal shock will not be included in this test.

4.6.8 Shock. In the nonoperating (stowed) condition, the test set shall be subjected to the shock test of MIL-T-21200, under "Shock, Equipment (not including transit case)," along its three principal axes. At the conclusion of this test, the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

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4.6.9 Vibration, operating. The test set shall be mounted so as to simulate service use (control panel horizontal) with the cover removed. The test set shall be operated in the automatic mode at the 0.5 ohm continuity check position with Shorting Adapter Assembly "A" (NAVORD Dwg 2837970) installed in place. The test set shall be recycled as necessary during the vibration test. The test set shall be subjected to the vibration tests defined in MIL-STD-167, Classification Type I. False indication or slipping of sequential stations shall be considered failure to meet the requirements of this specification. At the conclusion of this test, the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

4.6.10 Vibration, transportation. In the nonoperating (stowed) condition, the test set shall be subjected to the transportation vibration test of MIL-T-18404 (under "Torpedo, nonoperating, ready nonoperating, or ready") along its three principal axes. At the conclusion of this test the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

4.6.11 Marine environment. The test set shall be tested in a simulated service condition. The case covers shall be removed and set aside, the tray removed from the adapter case, and the power cord and Calibration Test Adapter Assembly (NAVORD Dwg 2837984) installed in their respective front panel connectors. The test set, prepared as above, shall be exposed for a period of 240 hours to sea atmosphere as experienced at dockside or on the weather deck of a ship. The test set shall be protected from environmental conditions in such a manner as to permit free circulation of air around it. At the conclusion of this test the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

4.6.12 Temperature and humidity cycling. The test set shall be tested in a simulated service condition. The case covers shall be removed and set aside, the tray removed from the adapter case, and the power cord and Calibration Test Adapter Assembly (NAVORD Dwg 2837984) installed in their respective front panel connectors. The test set shall be subjected to temperature and humidity cycling tests for Class 4 equipment called out in MIL-E-16400. At the conclusion of the test the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

4.6.13 Tropical environment. The test set shall be tested in a simulated service condition. The case covers shall be removed and set aside, the tray removed from the adapter case and the power cord and Calibration Test Adapter Assembly (NAVORD Dwg 2837984) installed in their respective front panel connectors. The test set shall be subjected to the Tropical Environment Test defined in MIL-T-18404. At the conclusion of the test, the test set shall

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be subjected to the operating and insulation resistance tests of 4.6.2 and 4.6.5 respectively.

4.6.14 Accelerated life test. The test set shall be subjected to the accelerated life test for Class 4 equipment described in MIL-E-16400. At the conclusion of the test, the test set shall be subjected to the operating and insulation resistance tests described in 4.6.2 and 4.6.5 respectively.

4.6.15 Packaging and packing. The packaging and packing of the cable assembly shall be examined to ensure that it complies fully with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The test set shall be packaged in accordance with Method III of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The test set shall be packaged in accordance with manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The test set, packaged as described in 5.1.1.1, shall be packed in a fiberboard box in accordance with PPP-B-636, class weather resistant."

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 100. Battens shall be utilized to eliminate the underhang.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

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5.3 Marking

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Torpedo Control Cable Electrical Test Set Mk 519 Mod 0 is intended for use in heated work areas on tenders and shore facilities to inspect and test torpedo control cables and certain associated items.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements

(a) Title, number, and date of this specification.

(b) Any exception to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction inspections.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.3.2	Certified inspection data	-

DIDs (Data Items Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project No. 6625-N617

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DOD-316

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Standardization Division (502)
Indian Head, MD 20640

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-T-82641(OS), Test Set, Electrical, Torpedo Control Cable, Mark 519 MOD 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N 0102-014-1801 C-25254

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